



F. Three Quick Brown Foxes Jump Over a Lazy Chicken

CPU: 1.00s

Memory: 2048MB

Three clever foxes each guard a side of a triangular field. This field is fenced on all sides, and within its boundary roams a cautious chicken. Every day, the chicken roams freely around the field, covering every point of the space within the triangle, yet never touching the fenced edges that form its boundary.

Eager to catch the chicken, the foxes devise a plan. First, each fox must find the perfect spot along their respective side of the triangle to take position. Then, they must identify a point inside the triangle to place a trap. Their goal is to create a setup where they can jump from their positions to this trap, reaching it at the same time to catch the chicken, while each fox covers the shortest possible distance at the same speed. Note that no two foxes will share the same side of the triangle. The foxes can't take positions on the vertices of the triangle.

The challenge is now clear: with only the lengths of the triangle's sides to guide them, the foxes must figure out the precise positions and shortest synchronized jump that will lead them to the trap and, finally, to the elusive chicken.

You will help the foxes figure out the shortest possible distance of the jump needed, given the lengths of the sides of the triangular field.

Input Description

The first line will contain a single integer T ($1 \leq T \leq 1000$). Each test case will have three integers X , Y , and Z ($1 \leq X, Y, Z \leq 1000$), denoting the lengths of three sides of the triangular field. It is guaranteed that the side lengths will make a valid triangle.

Output Description

For each test case, let the length of the shortest possible jump is L . You have to print the square of L as an irreducible fraction. An irreducible fraction is a fraction in which the

numerator and denominator are integers that have no common divisors other than 1.

Samples

Input	Output
2	32/1
57 13 60	1176/23
103 17 110	

ICPC PRELIMINARY DHAKA SITE 2024

Ends in
03:15:10

THE CONTEST IS RUNNING

SUBMIT

Language

C++

-O2 -static -std=gnu++17

SELECT SOURCE

SUBMIT

MY SUBMISSIONS



No submissions yet!